

Gunshot Residue Classification Report

Gunshot Residue Classification: Accuracy, Interpretability, and Failure Analysis.

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Slide 1 – Title

Thank you for joining us today. Our project is titled **Gunshot Residue Classification: Accuracy, Interpretability, and Failure Analysis**.

Our primary stakeholder is the **National Institute of Justice**, along with forensic laboratories and legal organizations that rely on accurate forensic evidence.

The purpose of this project is to evaluate whether machine learning can classify gunshot residue particles accurately and reliably, while also identifying where these models fail. This distinction is important because forensic models cannot be evaluated on accuracy alone. In legal settings, false positives may have serious consequences, so the model must also support interpretability, uncertainty recognition, and expert review.

Slide 2 – Stakeholders & Why This Matters (Background)

Gunshot residue analysis is important because it can influence criminal investigations and courtroom decisions. Traditionally, experts use (SEM/EDS) analysis to identify particles containing elements such as **lead, barium, and antimony**.

The challenge is that some environmental materials, like brake dust, fireworks, or industrial particles, can look chemically similar to GSR. That creates a risk of false positives, which is especially serious in legal settings.

Slide 3 – Research Question, Hypothesis & Predictions

Our research question was:

How accurately and reliably can machine learning models distinguish true gunshot residue particles from chemically similar non-GSR particles, and what are the limits of these models when generalizing to other similar datasets?

Our hypothesis was that machine learning models would perform very well when classic multi-element GSR signatures are present, especially combinations of lead, barium, and antimony.

We also predicted that single-element particles would be much harder to classify confidently, because a particle containing only lead, barium, or antimony does not provide enough chemical evidence on its own.

Lastly, we predicted that non-linear models such as XGBoost and Neural Networks will outperform linear models like Logistic Regression.

Slide 4 – Data Sources

We used two public forensic datasets.

The main dataset came from the Netherlands Forensic Institute, which included over 2.8 million particles measured across elemental composition columns. This was our primary modeling dataset.

We also used a NIST gunshot residue dataset as an external reference to support the labeling approach, although it was not merged into the training data. Later on, we used this dataset to test our models' abilities to distinguish between GSR and Non-GSR particles on a dataset from a different lab.

After cleaning and labeling, particles were classified into three groups: GSR, Non-GSR, and Ambiguous. Ambiguous particles were single-element classes such as lead, barium, antimony, strontium.

Slide 5 – Data Cleaning & Variable Selection

Before modeling, we removed ambiguous particles from the binary classification task because they could not be confidently labeled as either GSR or Non-GSR.

That left a final modeling dataset of around 2.3M particles, with about 47% GSR and 53% Non-GSR. The response variable was binary: GSR equals 1, and Non-GSR equals 0.

We also retained the `stub_id` so that particles from the same physical sample would not be split across training and testing. This group-based split was important because otherwise the model could learn sample-specific patterns instead of generalizable chemical patterns.

Slide 6 – Modeling Approach

We selected three models to intentionally span the interpretability-performance tradeoff: logistic regression, XGBoost, and a neural network.

We begin with logistic regression as our interpretable baseline, with transparent coefficients that can be explained in a forensic setting and a relatively low risk of overfitting.

XGBoost handles sparse, high-dimensional data effectively and provides built-in feature importance through a non-parametric approach.

The neural network offers the most flexibility, allowing us to learn complex, non-linear interactions without relying heavily on manual feature engineering.

All three models were developed and tested in parallel using the same underlying dataset and evaluation framework. This allowed us to directly compare performance across model types and understand how increasing complexity impacts results.

To ensure valid comparisons, we implemented a group-aware train/test split using `stub_id` to prevent data leakage, applied model-specific feature engineering, and removed any features that acted as label proxies.

Slide 7 – Key EDA Insights

Our EDA showed that the data is strongly non-linear.

- PCA, a linear method, only explains about 16.5% of the variance and shows heavy overlap between GSR and Non-GSR, meaning linear separation is limited.
- UMAP, which captures non-linear structure, shows much clearer clustering and achieves about a 94% kNN accuracy.

We also see that barium appears in both classes, especially near boundaries, which explains many false positives.

These findings guided our modeling approach and led us to test both linear and non-linear models.

Slide 8 – Logistic Regression - Interpretable Baseline Analysis

Logistic regression was used as the interpretable baseline because it provides transparent coefficients and offers a useful comparison point against the more flexible non-linear models.

The model used proportional elemental features, log transformations, and engineered ratio and interaction terms to better approximate the relationships observed during EDA.

With the full feature set, logistic regression achieved near-perfect performance, with ROC-AUC and PR-AUC values close to 1.0. However, when the core GSR elements—**lead, barium, and antimony**—were removed, performance dropped to approximately 0.90.

This decline indicates that logistic regression depends heavily on the strongest elemental signatures and has limited ability to capture more complex non-linear relationships on its own.

Overall, the results confirm its value as an explainable benchmark, while also supporting the need to evaluate more flexible models such as XGBoost and neural networks.

Slide 9 – Model Selection: XGBoost or Neural Net

The comparison between XGBoost and the neural network shows that both non-linear models perform at a very high level on the NFI test set.

Across the primary evaluation metrics, including accuracy, ROC-AUC, and PR-AUC, the two models are nearly indistinguishable. This supports the conclusion that the binary classification task is highly separable when evaluated within the same data source.

The more meaningful distinction is their behavior across operating thresholds. XGBoost is generally more conservative, producing fewer false positives in several threshold settings, while the neural network is more sensitive and more likely to classify borderline particles as GSR.

In a forensic context, this distinction matters because model selection is not only about overall accuracy, but also about the tradeoff between false positives and false negatives.

This comparison sets up the next analysis, where we examine whether the strong performance reflects reliable learning or possible overfitting, leakage, or uncertainty in ambiguous cases.

Slide 10 – Failure Mode Checks

Given the near-perfect performance, we evaluated whether the models are capturing true signal rather than overfitting or leakage.

The learning curves on the left show close alignment between training and validation performance for both models, indicating no evidence of overfitting.

On the right, feature ablation demonstrates that removing engineered features does not materially degrade performance, suggesting that results are not driven by leakage or overly engineered inputs.

Together, these findings indicate that the underlying signal in the data is strong.

However, strong aggregate performance does not necessarily reflect behavior on uncertain or ambiguous cases—particularly those lacking clear multi-element GSR signatures.

Slide 11 – Ambiguous Particle Results

The ambiguous particle test examines cases with only one potentially relevant element (antimony, lead, barium, or strontium), where the correct outcome should reflect uncertainty rather than a clear GSR/non-GSR label.

Both models behave cautiously overall, classifying most of these particles as non-GSR. The neural network is especially conservative, while XGBoost shows similar trends except for antimony, where more cases fall into the uncertain range and about 9.3% are predicted as GSR.

SHAP Animation

SHAP analysis suggests this is driven by a feature combining barium and antimony, which can mimic a partial GSR signal. Overall, the results highlight that single-element particles are handled conservatively, but antimony remains a key source of uncertainty—supporting the need for a “defer to expert” outcome when evidence is insufficient.

Slide 12 – Where Errors Happen: False Positive Analysis

For you as forensic professionals, the most critical error type is false positives — incorrectly identifying non-GSR particles as gunshot residue.

Logistic regression shows the highest false positive rate, reflecting its limitation in capturing the non-linear relationships that characterize GSR. In contrast, both XGBoost and the neural network achieve over 99.9% accuracy, indicating that the task is well-suited to non-linear models.

However, their error profiles differ in ways that matter operationally. XGBoost is more conservative, with the lowest false positive rate (0.07%), while the neural network is more aggressive, producing a higher rate (0.37%). Given the evidentiary risk of false positives, XGBoost’s behavior may better align with forensic requirements for caution and reliability.

Slide 13 – Where Errors Happen: False Positive Analysis Cont.

The false positives are not randomly distributed across particle types. They are concentrated in chemically similar non-GSR classes, especially the barium-heavy classes.

Both models most often misclassify **Barium–Aluminium** and **Barium–Calcium–Silicon** particles. XGBoost produces relatively few such errors (107 and 40 cases, respectively), while the neural network shows the same pattern at a larger scale.

This reflects a key chemical issue: **barium** is a core indicator of GSR, but it also appears in environmental and industrial particles. As a result, barium-heavy particles without the full **Lead–Barium–Antimony** signature can resemble true GSR and trigger false positives.

Other particle types, such as **Copper–Zinc** and **Titanium–Zinc–Gadolinium**, contribute only marginally to errors.

Overall, the findings highlight a central forensic challenge: while barium is highly informative, it is also the primary driver of false positive risk when not accompanied by the complete multi-element GSR profile.

Slide 14 – Testing on data from a different lab: NFI -> NIST

Cross-lab validation significantly changes how model performance should be interpreted.

While XGBoost and the neural network perform extremely well on NFI data, their performance does not generalize to the National Institute of Standards and Technology dataset. On this external data, false positive rates increase sharply — 54.0% for XGBoost and 73.7% for the neural network — with many errors made at high confidence.

This indicates the models are likely learning patterns tied to NFI-specific measurement conditions rather than fully generalizable gunshot residue chemistry.

The key implication is that, although promising for intra lab use, these models are not yet suitable for cross-laboratory deployment without additional validation, standardization, and domain adaptation.

Slide 15 – Recommendations and Next Steps

The main recommendation is that these models should not be treated as universally deployable without additional cross-lab validation.

Within the NFI dataset context, the models appear highly effective for the binary GSR classification task. However, the NIST dataset validation shows that they also learn instrument- and lab-specific measurement patterns, not fully generalizable GSR chemistry.

Future work should therefore focus on domain adaptation, especially methods that can learn instrument-invariant features across forensic labs.

A second recommendation is to avoid forced classification for single-element ambiguous particles. These cases should include a “defer to expert” outcome because they lack enough chemical evidence for confident automated classification.

Finally, broader standardization is needed. Differences in available elements between NFI and NIST datasets show that forensic labs need a shared minimal element panel to improve model transfer.

Overall, the models are promising within a controlled lab context, but production use across labs requires domain adaptation, uncertainty detection, and stronger forensic standardization.